VoiceOps 2026: ZentroMart Voice Support Agent

Name: NAME_PLACEHOLDER · **Repo:** REPO_PLACEHOLDER · **LLM:** `openai/gpt-oss-120b` on Groq (free tier) · **Speech:** Gnani STT + TTS only (`timbre-v2.5`, voice Kaveri)

Customer audio → Gnani STT (en-IN) → spoken numbers → digits →
Agent: query rewrite · hybrid RAG · tools · memory → Gnani TTS → Spoken reply + JSON log

1. What I built

Step 1: LLM only. One chat call with a system prompt that makes the model ZentroMart's AI support assistant. The prompt covers scope (orders, delivery, returns, refunds, payments, warranty), limits (never invent policy or promise compensation), a short speakable style, replying in the customer's language, and stating in the first reply that it is an AI assistant. It has no documents and no tools; this is the baseline.

Step 2: Grounding (RAG). The 21 KB files are split into one chunk per `[SECTION-ID]` section (84 sections), and each chunk keeps its `doc_type`, `status` and `effective_from`.

- **Hybrid retrieval:** BGE-small embeddings (fastembed/ONNX) and BM25 on unicode tokens, so the Hindi FAQ is searchable too, merged with reciprocal-rank fusion. Results are weighted by the KB-00.1 precedence order: policy > agent guide > help > FAQ > marketing.
- **Query rewriting:** a small model turns each message into a standalone English query, mapping customer words to document words ("send it back" → return, "nobody came to collect it" → reverse pickup). It also translates Hindi/Hinglish and resolves follow-ups.
- **Superseded policy:** `pol_returns_v3` stays searchable, because RET-3 still applies it to electronics delivered before 1 August 2026, but it is down-weighted and labelled.
- **Always in context:** the rules that govern every turn (STY-1/3/4, IDV-1..4, ESC-1/2/4, KB-00.1).
- **Grounding rule:** the model answers only from the given sections and tool results. When they don't cover the question, it says "I don't have information about that" and offers a colleague.
- **Source tracking:** each answer ends with the section IDs it used. These are logged with their text, together with any data rows used (e.g. `products.csv:ELE-0012`), and stripped before the reply is spoken.

Step 3: Conversation. The agent keeps the full history of each call and has function-calling tools:

- the dataset's own `lookup_order(order_id, verification_value)` and `create_ticket`;
- `lookup_product` (products.csv) and `check_pincode` (pincodes.csv);
- `date_calc` with a `business_day_range` operation (Mon–Sat, minus `holidays.csv`, today = 20 Oct 2026). Example: a card refund started on 17 Oct gives 24–27 Oct.

Product names in a message are matched to the catalog in code, so the right return category is applied without an extra LLM call. The code also enforces the IDV rules: one value at a time, at most one retry, and the value is never logged. The agent claims a handover only after `create_ticket` returns `created=True`. ESC-1 cases (damaged items over ₹10,000, an order the customer didn't place, legal threats, safety risks, a request for a human) are escalated straight away.

Step 4: Voice. `run.py --input <folder> --output <folder>` runs Gnani STT → agent → Gnani TTS on every clip and saves the reply as `.wav` and `.txt`, plus a `.json` with the transcript, sections and tool calls. Files named `<conv>_turnN` share one session.

- Replies follow STY-3: two or three sentences, amounts in rupees, dates with the year, and long IDs read in groups of four digits.
- An empty transcript gets "Could you please say it again?"; an LLM failure gets an apology and an offer of a colleague, never silence.
- Callers who speak Hindi or Hinglish get the reply spoken in `hi-IN` / `hi-en`.
- Test clips were synthesised with Gnani TTS in a separate "customer" voice (`make_test_audio.py`), since the dataset has no audio and no other people were recorded.

2. Step 1 vs Step 2 on the 52 practice questions

Scoring is strict. A reply must contain every `must_include` group, contain no `must_not_include` phrase, **and** pass an LLM judge (`gpt-oss-20b`), which checks the expected decision and the reference answer's key facts. Retrieval is scored against `gold_evidence`. Both steps use the same model.

Metric	Step 1 (prompt only)	Step 2 (RAG + tools)
Correct: answer-key rules + LLM judge	**15.4% (8/52)**	**59.6% (31/52)**
Answer-key rules only (must / must-not include)	**32.7% (17/52)**	**80.8% (42/52)**
After my manual review of judge rejections*	–	**71.2% (37/52)**
Retrieval: a gold section in the context (hit@6)	–	**93.6% (44/47)**
Retrieval: recall@6 of gold sections	–	**86.2%**
Gold sections cited by the answer	–	**70.2%**
Step 2 by question language	**English 64% (21/33) · Hindi 50% (4/8) · Hinglish 55% (6/11)**	

* 11 replies passed the answer-key rules but were rejected by the judge. 6 of them state exactly the reference's facts: refund ranges such as "24–27 October 2026" in PQ0022, 27, 28, 52 and 59, and "7 days" in PQ0058. The judge rejected them only for not spelling out "Monday to Saturday, excluding holidays", or for extra correct detail. I report the strict number as the headline.

What I learned

- **Without documents the model guesses confidently.** Step 1 gave the common "5–7 days" for wallet refunds, which policy says are instant (within 2 hours). It kept the old 10-day electronics window (now 7 days) and made up "9 am–9 pm" delivery slots. It also flatly answered questions no document covers, such as store pickup or a student discount, instead of saying "I don't know". It was right only on questions that general knowledge happens to cover.
- **Grounding fixes most of this.** Correct answers rose from 15% to 60% under the strict judge (33% → 81% on the answer key alone), and the right section was in the context 94% of the time. Most remaining errors are reasoning over the retrieved text, not retrieval.
- **Date maths must be done in code.** Before the `business_day_range` tool, the model miscounted refund windows across the 19 October warehouse holiday. With it, every refund-date answer had the exact range.
- **The judge needs checking too.** A small judge model is stricter than the answer key on wording. Reading its rejections stopped me from "fixing" answers that were already right.

3. Conversations (Step 3) and voice (Step 4)

Practice conversations	Score
Turns correct (rules + judge)	64.3% (9/14)
Expected tool calls made	71.4% (10/14)
Clarify turns / escalate turns correct	100% / 100%
Answer turns correct	29% (2/7)

3 of 23 conversations (14 turns) fit in Groq's free 200k-tokens/day quota after the question runs. The evaluation is resumable (`eval.py --resume`) and continues from turn 15 on the next run.

Voice run (22 Gnani TTS clips)	Result
Clips with a transcript, reply and spoken reply	22/22
Conversations replayed in one session	3 (14 turns)
Hinglish transcribed and answered	yes
Speech services used	Gnani only

Examples are in `sample_outputs/`, e.g. PC002: greeting → order ID → verification → damaged TV over ₹10,000 → ticket to Priority Resolution.

4. What went wrong and what I would fix

Problem (practice set)	Example	Status / fix
STT writes numbers as words ("eight nine seven eight...", "do hazaar nao sau unchas"), so a verification failed in a voice conversation	PC004: phone digits 2949 heard as Hindi words	**Fixed:** `numbers.py` converts English and romanised Hindi number words to digits before the agent sees them (IDs, pincodes, years, "4533"). Tested on all 22 real transcripts.
Festive-sale exception missed (RET-8: orders placed 1–15 Oct get 30 days)	PQ0049, 0054, 0056: said 15 days for a 5 Oct sale order	Open. Next: a `return_window(category, order_date, delivery_date)` tool, so RET-2/3/8 are applied in code like the date maths.
Over-cautious "I don't know" after the strict grounding rule	PQ0034 Dubai, PQ0024 kurta, PC004 jeans (in Hindi)	Partly fixed with catalog matching in code. Next: retry retrieval with the raw question before giving up.
Asked for the order ID when a general answer was possible	PQ0042 "already shipped, can I stop it?", PQ0048	Prompt rule added; still inconsistent. Next: give the policy first, then offer to check the order.
Replied in English to Hindi/Hinglish in some answers, despite STY-4	PQ0013, PQ0027, PQ0052	Next: check the reply language against `detect_language` and regenerate once if it doesn't match.
Escalated one turn early when the order record already showed a damage claim	PC002 turn 3	Acceptable for the customer, but costs a turn. Next: answer the verified-order turn, then escalate.
Retrieval misses (3/47): RET-4 "opened beauty items" and RET-2 for a kurta	PQ0013, PQ0016, PQ0024	Next: a cross-encoder re-ranker, plus category keywords from the catalog match in the query.
Free-tier limits: Groq allows 200k tokens/day, but a full evaluation needs about 1M	First full run stopped at the daily limit	**Fixed:** prompt 3.3k → 2.8k tokens, low reasoning effort, small model for rewriting and judging, wait on per-minute limits, clean stop on the daily limit, resumable runs.

5. Bonus work

- **Hindi & Hinglish end to end:** unicode BM25 over the Hindi FAQ, query translation, replies in the caller's language, and Hindi TTS for Hindi replies.
- **Spoken-number normalisation** for STT output, covering English and Hindi number words, digit-by-digit IDs and years.
- **Deterministic helpers:** business-day ranges with holidays, catalog matching, and IDV retry limits with masked logs, so the LLM is not trusted with arithmetic or security.
- **Evaluation tooling:** answer-key rules plus an LLM judge, retrieval hit/recall, tool-call matching, an ASR-text mode (`--use-asr`), resumable runs, and `rejudge.py` to re-score saved answers without re-running the agent.
- **Reproducibility:** `run_all.bat` runs the whole pipeline in one click within free-tier limits; `app.py` is a Gradio microphone demo that shows the sections and tool calls behind each answer.